

Predicting Student Academic Success Using Machine Learning

Project Summary Report

CPE393: Introduction to Data Science with Python | Summer Semester 2026 | KMUTT

Group: Taiba

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GitHub Repository: https://github.com/azzedineh27/Final_project_taiba

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2. Problem Definition and Goals

Schools generally only find out that a student is struggling once final grades are already in, by which point it is too late to intervene. The goal of this project is to test whether a student's background, family situation, and behaviour during the school year already carry enough signal to flag at-risk students earlier, without waiting for the final exam.

We frame this as a binary classification problem. The target variable, pass, is derived from the final grade G3 (0-20 scale) using the standard Portuguese grading convention: a student with G3 of 10 or above is labelled as a pass (1), and a student below 10 is labelled as a fail (0). Everything else in the dataset -- demographics, family background, study habits, school support, and social behaviour -- is treated as a candidate predictor.

Concretely, the project answers three questions: (1) How well can non-grade information alone predict pass/fail? (2) Which factors matter most? (3) What happens to that prediction once in-course grades are added, and what does that reveal about how such a model should and should not be used in practice? Question 3 turned out to be one of the most informative parts of the analysis and is discussed at length in Sections 8 and 9.

This matters because a model that works reasonably well using only background and behavioural data could, in principle, support a school's decision about who to offer extra tutoring or counselling to at the start of a term, rather than after a student has already fallen behind. We treat this as a decision-support idea to be discussed carefully, not as a finished product -- see Section 9 for the limits of that claim.

2.1 Why classification rather than regression

The final grade G3 is itself a numeric score (0-20), so predicting it directly would be a regression problem. The group chose to convert it into a binary pass/fail label instead, for two reasons. First, the practical decision a school actually needs to make is closer to a yes/no question -- "does this student need extra support or not" -- rather than an exact numeric grade forecast, so a classification framing matches the real decision more directly. Second, a pass/fail framing makes model evaluation more directly interpretable through precision, recall, and confusion matrices, which map naturally onto the real cost of the two possible mistakes: missing a student who needed help (a false "pass"), versus flagging a student who did not need it (a false "fail").

2.2 Scope of this report

This report covers one subject (Mathematics) from one dataset. A companion dataset exists for a Portuguese-language course from the same schools; combining the two was considered but left for future work (Section 10.4) so that this report could focus on doing one analysis thoroughly rather than two analyses superficially.

3. Dataset Description

Source: UCI Machine Learning Repository / Kaggle mirror, "Student Performance Data Set", collected by Cortez and Silva (2008) from two secondary schools in Portugal. This report uses the Mathematics course subset, distributed as studentmat.csv.

License / usage condition: the dataset is publicly released for research and educational use and is commonly redistributed on Kaggle and the UCI repository. No personally identifying information (names, addresses) is included.

Size and format: 395 rows (one row per student) and 33 columns, provided as a comma-separated CSV file. There is no separate test file; the group performed its own train-test split (Section 6).

Feature groups: demographic (age, sex, address type), family background (parents' education and jobs, family size, cohabitation status), school-related (travel time, study time, past failures, school and family educational support, extra paid classes), lifestyle (free time, going out, weekday/weekend alcohol consumption, health), and academic history (G1 and G2, the first- and second-period grades). The target label used for classification, pass, is engineered from G3, the final grade.

Table 0 below lists every column exactly as it appears in the raw file, grouped the same way, so that any reader of this report can look up what a given feature means without needing to consult the original UCI documentation separately.

3.1 Variable dictionary

Column	Meaning	Type / range
school	Student's school	Categorical: GP, MS
sex	Student's sex	Categorical: F, M
age	Student's age	Numeric: 15-22
address	Home address type	Categorical: U (urban), R (rural)
famsize	Family size	Categorical: LE3 (≤ 3), GT3 (> 3)
Pstatus	Parents' cohabitation status	Categorical: T (together), A (apart)
Medu	Mother's education	Numeric: 0 (none) - 4 (higher)
Fedu	Father's education	Numeric: 0 (none) - 4 (higher)
Mjob	Mother's job	Categorical: teacher, health, services, at home, other
Fjob	Father's job	Categorical: teacher, health, services, at home, other
reason	Reason for choosing this school	Categorical: home, reputation, course, other
guardian	Student's guardian	Categorical: mother, father, other

Column	Meaning	Type / range
traveltime	Home-to-school travel time	Numeric: 1 (< 15 min) - 4 (> 1 hour)
studytime	Weekly study time	Numeric: 1 (< 2 h) - 4 (> 10 h)
failures	Number of past class failures	Numeric: 0-3 (capped)
schoolsup	Extra educational support from school	Categorical: yes, no
famsup	Extra educational support from family	Categorical: yes, no
paid	Extra paid classes for the subject	Categorical: yes, no
activities	Extra-curricular activities	Categorical: yes, no
nursery	Attended nursery school	Categorical: yes, no
higher	Wants to pursue higher education	Categorical: yes, no
internet	Internet access at home	Categorical: yes, no
romantic	In a romantic relationship	Categorical: yes, no
famrel	Quality of family relationships	Numeric: 1 (very bad) - 5 (excellent)

Column	Meaning	Type / range
freetime	Free time after school	Numeric: 1 (very low) - 5 (very high)
goout	Going out with friends	Numeric: 1 (very low) - 5 (very high)
Dalc	Workday alcohol consumption	Numeric: 1 (very low) - 5 (very high)

Walc	Weekend alcohol consumption	Numeric: 1 (very low) - 5 (very high)
health	Current health status	Numeric: 1 (very bad) - 5 (very good)
absences	Number of school absences	Numeric: 0-75
G1	First-period grade	Numeric: 0-20
G2	Second-period grade	Numeric: 0-20
G3	Final grade (target source)	Numeric: 0-20
pass (engineered)	1 if G3 >= 10, else 0	Binary: 0 (fail), 1 (pass)

Table 0. Full variable dictionary for studentmat.csv, grouped by demographic, family, school-related, lifestyle, and academic-history features.

Limitations: the sample is small (395 students), drawn from a single subject, two schools, and one region and time period (Portugal, 2005-2006). Several columns (alcohol consumption, family relationship quality, free time) are self-reported survey answers and may carry reporting bias -- a student's own sense of "very good" health or "excellent" family relations is subjective and not independently verified. There is also no information on teacher quality, classroom size, or curriculum changes between the two schools, all of which could plausibly affect G3 but are not captured anywhere in this dataset. These limitations restrict how far conclusions can be generalised beyond this specific population, and are revisited in Sections 9 and 10.

4. Preprocessing and Cleaning

Before any modelling, the dataset was checked against a standard cleaning checklist: missing values, duplicate rows, incorrect data types, outliers, and unnecessary columns. Each check and the resulting decision is reported below rather than assumed.

4.1 Missing values

A cell-by-cell null check (`df.isnull().sum()`) found zero missing values across all 395 rows and 33 columns. No imputation was required.

4.2 Duplicate rows

An exact row-duplication check (`df.duplicated()`) found zero duplicate rows, so no rows were removed on this basis.

4.3 Data types

Numeric columns (age, grades, study time, failures, and the lifestyle rating columns) were already stored as integers. Categorical columns (school, sex, parents' jobs, reason for choosing the school, guardian, and several yes/no columns) were stored as text, which is appropriate at this stage; they were converted to numeric form during feature engineering (Section 6), not during cleaning, so that the cleaning step stayed focused on correctness rather than model-readiness.

4.4 Outliers

Boxplots were produced for the key numeric columns (age, absences, G1, G2, G3, studytime, failures), shown in Figure 1.

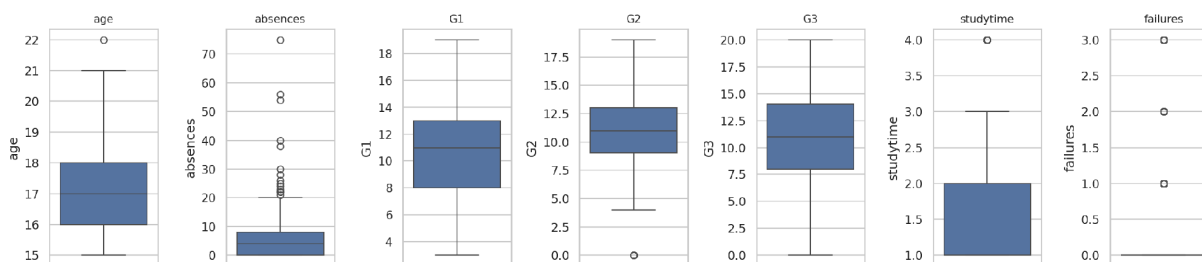


Figure 1. Boxplots of key numeric columns used to check for outliers.

absences shows a right-skewed tail: most students have few absences, but a small number have 20 or more. A handful of students have G3 equal to zero, which in this dataset corresponds to students who did not sit the final exam rather than a data-entry error. None of the values found are outside the physically plausible range (no negative ages, no absence counts below zero, no grades above the 20-point maximum), so these points were kept rather than removed -- they represent real, informative cases, and this project specifically cares about the tail of at-risk students.

4.5 Unnecessary columns

A check for constant columns (`df[c].nunique() == 1`) found none, and the dataset contains no student ID or other non-informative identifier column. No columns were dropped for redundancy during cleaning. (Two columns, G1 and G2, were later excluded from the main feature set for a modelling reason discussed in Section 6 and 8, not a data-quality reason.)

4.6 Cleaning summary

Check	Method	Result	Action taken
Missing values	<code>df.isnull().sum()</code>	0 missing cells	None required
Duplicate rows	<code>df.duplicated().sum()</code>	0 duplicates	None required
Data types	<code>df.dtypes</code>	All correct for their content	None required
Outliers	Boxplots (Fig. 1)	Some extreme but plausible values in absences, G3	Kept -- valid, informative data points
Constant / ID columns	<code>df[c].nunique() == 1</code> check	None found	None required

Table 0c. Summary of the five-point cleaning checklist applied to *studentmat.csv*.

5. Data Visualization and Exploratory Data Analysis

Descriptive statistics (`df.describe()`) show the numeric ranges at a glance: age 15-22 (mean 16.7), study time coded 1-4 (mean 2.0, i.e. between 2 and 5 hours per week on average), failures 0-3 (mean 0.33, i.e. most students have none), and absences 0-75 (mean 5.7, heavily right-skewed as shown above). The full summary is reproduced in Table below for reference, since several of these numbers are referred back to later in the report.

Statistic	age	Med u	Fedu	studyt ime	failur es	absen ces	G1	G2	G3
mean	16.70	2.75	2.52	2.04	0.33	5.71	10.91	10.71	10.42
std	1.28	1.09	1.09	0.84	0.74	8.00	3.32	3.76	4.58

min	15	0	0	1	0	0	3	0	0
50% (median)	17	3	2	2	0	4	11	11	11
max	22	4	4	4	3	75	19	19	20

Table 0b. Selected descriptive statistics for the main numeric columns.

5.1 Class balance

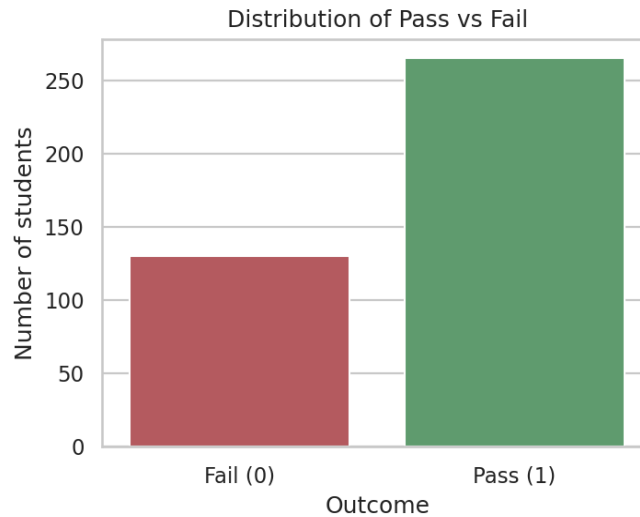


Figure 2. Distribution of the pass/fail target variable.

67.1% of students pass and 32.9% fail -- a moderate class imbalance. This is an important observation for evaluation: a model that always predicts "pass" would already score roughly 67% accuracy without learning anything, which is why Section 8 reports precision, recall, and F1-score rather than accuracy alone. An imbalance this size is not extreme enough to require resampling techniques such as SMOTE, but it is large enough that it must be kept in mind when reading any single accuracy figure, which is why every result in this report is presented alongside precision, recall, and F1-score rather than in isolation.

5.2 Distribution of the final grade

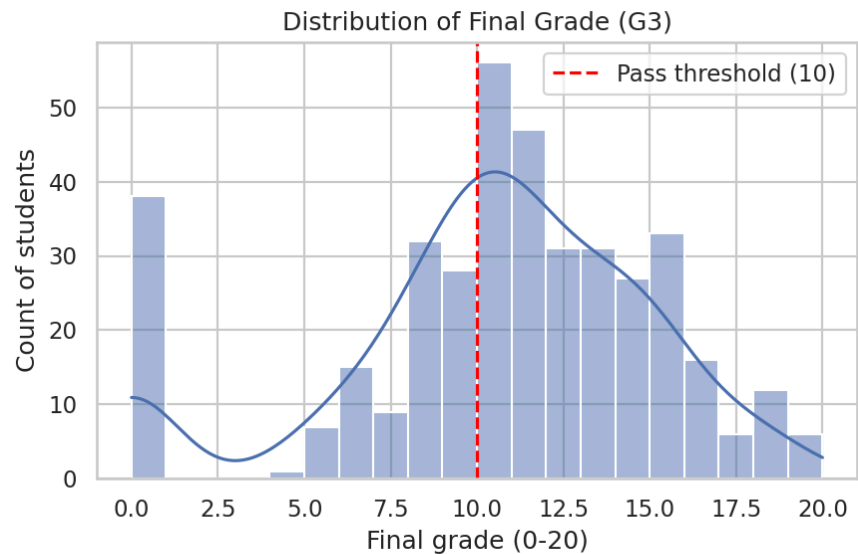


Figure 3. Distribution of the final grade (G3), with the pass threshold marked.

Grades are roughly bell-shaped with a spike of zeros (non-participation) and a visible cluster of students close to the pass boundary of 10. These "close-call" students are precisely the group an early-warning system would be most useful for.

5.3 Study time and absences versus outcome

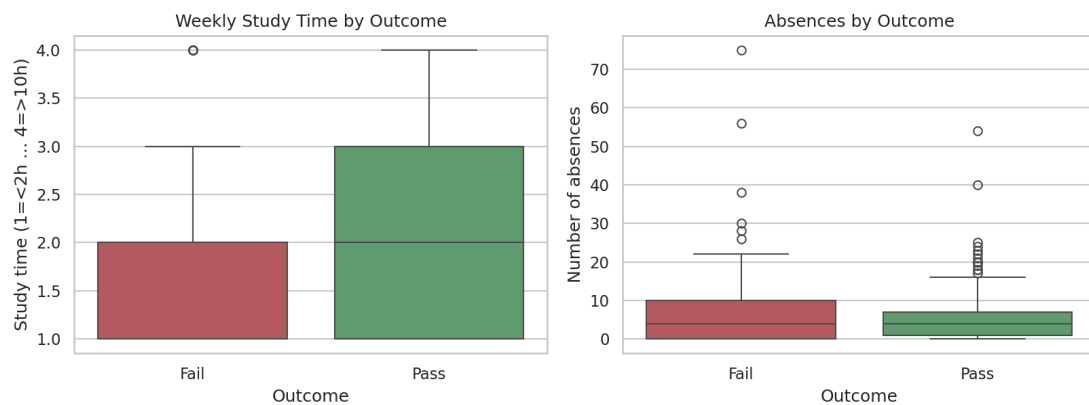


Figure 4. Weekly study time and number of absences, split by pass/fail outcome.

Students who pass report slightly higher median study time, and the failing group shows a somewhat wider spread of absences. The effect is real but modest when looking at either feature in isolation -- no single feature cleanly separates the two groups, which is the motivation for combining many features in a model rather than using a simple rule of thumb.

5.4 Correlation analysis

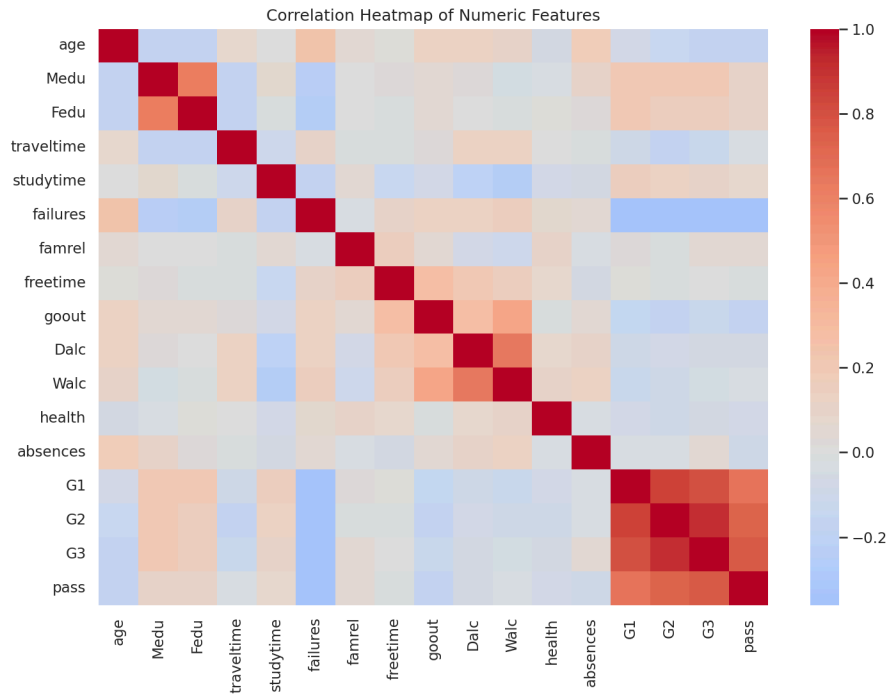


Figure 5. Correlation heatmap of all numeric features.

Ranking numeric features by their correlation with G3, the two strongest are G2 ($r = 0.90$) and G1 ($r = 0.80$) -- unsurprising, since they are earlier measurements of the same underlying quantity. Among the non-grade features, failures has the strongest relationship with the final grade ($r = -0.36$, i.e. more past failures is associated with a lower final grade), followed by mother's and father's education (Medu $r = 0.22$, Fedu $r = 0.15$) with a positive association, and age ($r = -0.16$) and going out with friends, goout, ($r = -0.13$) with a negative association. This early observation that G1/G2 dominate the correlation table foreshadows the leakage discussion in Section 8.

5.5 Categorical relationships

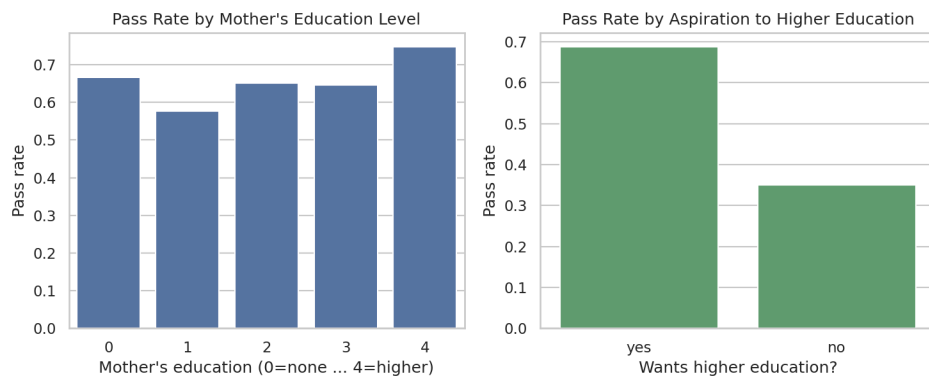


Figure 6. Pass rate by mother's education level and by aspiration to pursue higher education.

Pass rate rises fairly steadily with mother's education level, and students who report wanting to pursue higher education pass at a clearly higher rate than those who do not. Both are plausible proxies for home support and motivation, and both appear among the more informative features used later in modelling.

6. Feature Engineering

Four steps were applied to turn the cleaned table into a model-ready feature matrix.

6.1 Target variable

The binary column pass was created from G3 using the rule $\text{pass} = 1$ if $G3 \geq 10$, else 0, as described in Section 2.

6.2 Encoding categorical variables

Binary categorical columns (13 columns, e.g. sex, internet, schoolsup, romantic) were mapped to 0/1 codes, which is safe because there is no ordering ambiguity with only two categories. Multi-category nominal columns with no natural order (Mjob, Fjob, reason, guardian) were one-hot encoded using `pandas.get_dummies`, so that, for example, a mother's job of "teacher" versus "health" versus "at_home" does not get falsely represented as a numeric ranking. After encoding, the feature matrix has 39 columns (compared with 30 usable raw predictor columns), reflecting the extra one-hot columns created for the nominal variables.

6.3 Feature scaling

The 13 numeric columns (age, parents' education, travel time, study time, failures, family relationship quality, free time, going out, weekday/weekend alcohol use, health, absences) were standardised using scikit-learn's `StandardScaler`, which subtracts the mean and divides by the standard deviation so every numeric feature ends up with mean 0 and standard deviation 1. This matters for Logistic Regression, which computes a weighted sum of features and would otherwise let large-range columns such as absences (0-75) dominate over small-range columns such as age (15-22) purely because of units, not real importance. Tree-based models do not need scaling, but scaling was applied uniformly for a simpler, consistent pipeline. The scaler was fit only on the training set and then applied to the test set, to avoid leaking test-set statistics into preprocessing.

6.4 Feature selection: excluding G1 and G2 from the main model

G1 and G2 (first- and second-period grades in the same course) were deliberately excluded from the main feature set, even though they are numeric and available in the data. The reasoning, and a quantified demonstration of why this choice matters, is presented in Section 8.

6.5 Alternative approaches considered

Two alternatives were considered and rejected. First, ordinal integer-coding of the nominal job/reason/guardian columns (e.g. teacher=0, health=1, services=2, ...) was considered as a simpler alternative to one-hot encoding, but rejected because it would impose a false numeric ordering on categories that have none, which could mislead Logistic Regression in particular. Second, dropping low-variance lifestyle columns (such as famrel or nursery) before modelling was considered to reduce dimensionality, but rejected because the Random Forest feature-importance results (Section 8.2)

confirmed several of these columns do carry a measurable, sensible signal, so removing them in advance would have thrown away information without a clear justification.

7. Machine Learning Model Training

Three classification algorithms were trained and compared, as proposed at the start of the project.

7.1 Train-test split

The data was split 80% train / 20% test using scikit-learn's `train_test_split`, with `random_state=42` fixed for reproducibility and `stratify=y` so that both the training and test sets keep approximately the same 67/33 pass/fail ratio as the full dataset -- important given the moderate class imbalance and the fairly small sample size (395 rows, so 79 rows held out for testing).

7.2 Algorithms

- Logistic Regression: learns a weighted sum of the input features and passes it through the sigmoid function to produce a pass probability, which is then thresholded at 0.5. It is simple, fast, and its learned weights are directly interpretable in terms of direction of effect.
- Decision Tree: learns a sequence of yes/no splits on feature values (for example, "is failures greater than 0?") that progressively separate students into more pass-heavy or fail-heavy groups. Tree depth was capped at 5 to reduce the risk of the tree memorising the training data (overfitting), which is a particular risk with only 316 training rows.
- Random Forest: trains an ensemble of 200 decision trees, each on a randomly resampled subset of rows and a random subset of features, and combines their votes. Averaging over many trees typically reduces overfitting compared with a single tree and gives a more stable prediction. Depth was capped at 6 per tree.

All three models used `random_state=42` so that results are reproducible.

7.3 Hyperparameters used

Rather than leaving every model at scikit-learn's defaults, a small number of hyperparameters were deliberately set, mainly to control overfitting given the small training set (316 rows). Table below lists the values used and the reasoning behind each.

Model	Hyperparameter	Value	Reason
Logistic Regression	<code>max_iter</code>	1000	Default of 100 sometimes fails to converge with 39 features; raised to guarantee convergence.
Decision Tree	<code>max_depth</code>	5	An unrestricted tree can grow one leaf per training row and memorise the data; capping depth limits overfitting.
Random Forest	<code>n_estimators</code>	200	Enough trees for the ensemble average to stabilise without excessive training time.
Random Forest	<code>max_depth</code>	6	Same overfitting concern as the single Decision Tree, applied to each tree in the forest.
All models	<code>random_state</code>	42	Fixes all internal randomness so the notebook produces identical results on every run.

Table 3. Hyperparameters explicitly set for each model, and the reasoning behind each choice.

No exhaustive hyperparameter search (such as grid search or random search) was performed. Given the small dataset, an aggressive search would itself risk overfitting to this particular 80/20 split; a future iteration of this project using cross-validation (Section 10.4) would be a more appropriate setting in which to tune hyperparameters properly.

8. Evaluation Metrics and Results

Because the classes are imbalanced (67% pass / 33% fail), accuracy alone was judged insufficient: a model that always predicts "pass" would already score close to 67% without learning anything useful. Accuracy, precision, recall, and F1-score were therefore all reported, together with the confusion matrix for each model.

8.1 Main results (background and behavioural features only, no G1/G2)

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.658	0.717	0.811	0.761
Decision Tree	0.684	0.719	0.868	0.786
Random Forest	0.658	0.691	0.887	0.777

Table 1. Test-set performance of the three models using only background and behavioural features.

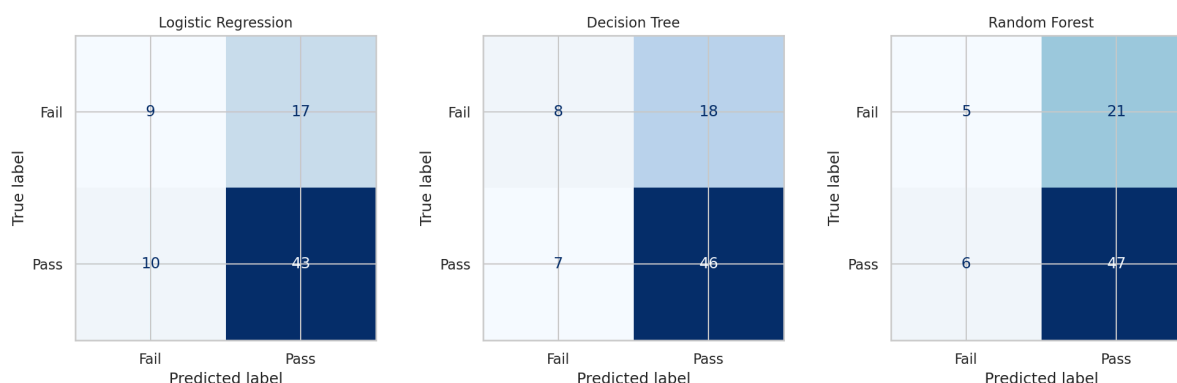


Figure 7. Confusion matrices for Logistic Regression, Decision Tree, and Random Forest.

All three models land in a similar range: roughly 66-68% accuracy and 76-79% F1-score on the pass class, clearly ahead of the ~67% a model would get by blindly predicting "pass" for everyone, but far from a strong result. The Decision Tree has the best accuracy (0.684) of the three, while Random Forest has the highest recall on the pass class (0.887). Because the practical goal of this project is to catch at-risk (failing) students early, the more relevant number in each confusion matrix is the top-left cell -- fail correctly predicted as fail -- and on that count all three models still miss a substantial fraction of failing students, which we report honestly rather than only citing the headline accuracy figure.

Looking at each model individually: Logistic Regression correctly identifies 9 of the 26 failing students in the test set (roughly one third), and misclassifies the rest as passes -- consistent with it being the simplest, most linear of the three models, unable to capture interaction effects such as "low study time combined

with high absences." The Decision Tree improves slightly on this, correctly catching 8 of 26 fails while also achieving the best overall accuracy, but a single tree of depth 5 is still a fairly blunt instrument on 316 training rows. Random Forest, despite having the same accuracy as Logistic Regression, catches the fewest failing students correctly (5 of 26) while being the most aggressive at predicting "pass" overall -- a useful reminder that a model can look identical on accuracy while behaving quite differently underneath, which is precisely why a single metric is not enough to compare models for this task.

Taken together, these results suggest that background and behavioural information alone captures a real but limited amount of signal about whether a student will pass. None of the three models would be reliable enough, on their own, to justify a high-stakes decision about an individual student; at best, they could contribute one input among several to a human counsellor's judgement.

8.2 Feature importance

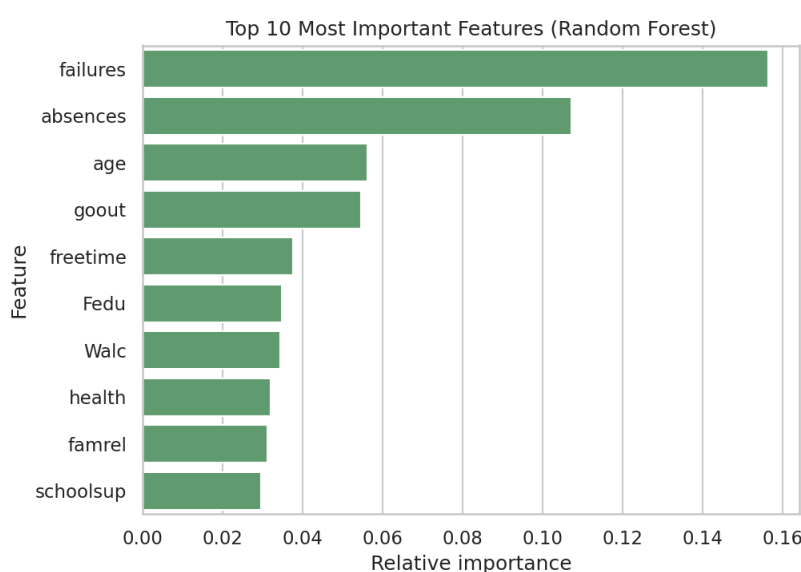


Figure 8. Top 10 most important features according to the Random Forest model.

The Random Forest's built-in feature importance ranks failures, absences, age, and goout (going out with friends) among the most influential predictors, followed by freetime and father's education (Fedu). These are all plausible, human-understandable drivers of academic outcome, which is a useful sanity check that the model is learning sensible patterns rather than an artifact of the data.

8.3 Data leakage case study: adding G1 and G2

Data leakage occurs when a model has access, directly or indirectly, to information that would not realistically be available at the time a prediction is needed, making performance look better than it would in real use. G1 and G2 are grades from earlier in the same course. Including them is not the classic "test data used during training" leakage, but it is closely related -- often called target leakage -- because G1 and G2 are so strongly correlated with G3 (Section 5.4: $r = 0.80$ and 0.90 respectively) that a model using them is largely restating a grade the student already has, rather than learning general risk factors from background and behaviour.

Model	Accuracy (no G1/G2)	F1 (no G1/G2)	Accuracy (with G1/G2)	F1 (with G1/G2)
Logistic Regression	0.658	0.761	0.886	0.911
Decision Tree	0.684	0.786	0.848	0.875
Random Forest	0.658	0.777	0.873	0.900

Table 2. Effect of including G1 and G2 as features. Accuracy rises by roughly 19-23 percentage points.

Accuracy rises from roughly 66-68% to 85-89% once G1 and G2 are added -- a dramatic change that would be easy to over-report as "our model reaches 89% accuracy" without qualification. Whether including G1/G2 is appropriate depends entirely on the intended use: for an early-in-the-year warning system that has to work before any in-course grades exist, G1 and G2 should be excluded, which is why they were excluded from the main model in Section 7-8.1. For a later, mid-year check-in after G2 already exists, including them would be legitimate and arguably the more useful model. Reporting both versions, and being explicit about which feature set produced which number, is the responsible way to present this result, and is revisited in Section 9.

9. Responsible AI Practices

9.1 Personal and sensitive information

The dataset includes family structure, parents' occupation and education, and weekday/weekend alcohol consumption. Although the data is anonymised (no names or student IDs), these are still sensitive attributes; predictions built on them could be misused to profile or label individual students if the underlying data or model outputs were ever linked back to real identities. Alcohol-consumption questions in particular touch on behaviour that is legally and socially sensitive for minors in most jurisdictions, and the fact that the survey was self-reported rather than independently observed does not remove that sensitivity -- if anything, it raises a separate question of whether students felt able to answer honestly, which is itself a limitation worth stating plainly rather than glossing over.

9.2 Bias, fairness, and source reliability

The data comes from two schools in one region of Portugal, collected in 2005-2006. Patterns the model has learned, including the apparent influence of parental education and address type (urban/rural), reflect the social and economic conditions of that specific population and time period, and may not transfer fairly to other countries, school systems, or years. Because some features (address, parents' job, parents' education) are correlated with socioeconomic status, using a model like this to make real decisions carries a risk of reinforcing existing inequities if it is not paired with human judgement and oversight.

9.3 Data leakage and honest reporting

As detailed in Section 8.3, the group deliberately evaluated the model both with and without G1/G2 and is reporting both sets of numbers rather than only the higher-looking, leakage-inflated figures. This is treated as a required part of honest reporting, not an optional detail.

9.4 Metric choice under class imbalance

Because the pass/fail split is imbalanced (67/33), accuracy alone is not sufficient. Precision, recall, and F1-score were reported for all models, with particular attention paid to how well each model catches the minority, fail, class, since that is the group an early-intervention system exists to help.

9.5 Real-world misuse risk

If deployed carelessly, a model like this could be used to label a student as "high risk" in a way that reduces the opportunities or attention given to them, becoming a self-fulfilling prophecy, rather than being used, as intended, to offer extra support. Given that even the best-performing model here still misses a meaningful share of at-risk students, and that some predictors correlate with socioeconomic background, this model should be treated strictly as a decision-support signal that supplements a teacher's judgement -- never as an automated or sole basis for decisions that affect a student's opportunities.

9.6 Checklist summary

Table below maps each item on the assignment's Responsible AI checklist directly to where it is addressed in this report, so that a reader (or grader) can verify each point was actually considered rather than only asserted.

Checklist item	Where addressed
Personal, sensitive, or biased information in the dataset	Section 9.1
Honest reporting of limitations	Sections 3, 8.3, 9.2, 10.3
Avoiding data leakage	Sections 6.4, 8.3
Metrics matching the task and class balance	Sections 5.1, 8.1
Real-world risks of misuse	Section 9.5
AI tool usage disclosure	Section 9.7

Table 4. Cross-reference between the Responsible AI checklist and the sections of this report.

9.7 Reproducibility

Every step in this report -- the cleaning decisions, the train-test split, and all three models -- uses a fixed `random_state` of 42, so any reader who runs the accompanying notebook on the same `studentmat.csv` file should obtain the exact same numbers reported in Tables 1 and 2. The full environment used was Python 3 with pandas, NumPy, Matplotlib, Seaborn, and scikit-learn; version numbers are listed in the notebook's first cell and in the accompanying `requirements.txt`. Reproducibility is treated here as part of responsible reporting: a result that cannot be independently re-run and checked is much harder to trust or to challenge.

9.8 Disclosure of AI tool use

Generative AI was used as a learning and support tool during this project. Specifically, it was used to:

- Explain machine learning concepts such as feature engineering, data cleaning, data leakage, evaluation metrics (accuracy, precision, recall, and F1-score), and model interpretation.
- Help understand Python code, Pandas operations, and Scikit-learn functions.
- Assist with debugging programming errors (for example, reading CSV files, handling encoding issues, and resolving Python exceptions).
- Suggest improvements to the structure and clarity of the notebook and report.

10. Conclusion and Future Work

10.1 Main findings

- Using only background and behavioural features (no in-course grades), all three models predict pass/fail with 66-68% accuracy and 76-79% F1-score on the pass class -- better than a naive always-"pass" baseline, but with real room for error.
- failures, absences, age, and goout are consistently among the most influential features, all of which are plausible and human-interpretable.
- Including G1 and G2 sharply raises apparent performance (to 85-89% accuracy), illustrating a leakage-style risk that must be disclosed rather than used to inflate headline results.

10.2 What the group learned

Beyond the technical workflow, the most useful lesson from this project was that a single accuracy number is not a complete answer -- the same dataset can be made to look far more or less impressive depending on which features are allowed in, and a responsible report has to make that choice explicit rather than choosing whichever number looks best.

10.3 Limitations

- Small dataset: 395 students, one subject, two schools, one region and era, which limits how far these findings generalise.
- Several features are self-reported survey answers, which can carry reporting bias (for example, alcohol consumption).
- No external validation was possible on a different school, subject, or year.

10.4 Future work

- Evaluate with cross-validation instead of a single train-test split, to get more stable performance estimates given the small sample size.
- Compare against the companion Portuguese-language-course dataset (student-por.csv) for a larger combined sample.
- Examine fairness metrics across subgroups (for example, by address type or sex) before considering any real deployment.
- Try additional models, such as gradient boosting, and tune hyperparameters more systematically.

References

Cortez, P., & Silva, A. (2008). Using data mining to predict secondary school student performance. Proceedings of 5th Annual Future Business Technology Conference.

Dataset: Student Performance Data Set, UCI Machine Learning Repository (Mathematics course subset, studentmat.csv).

Libraries used: pandas, NumPy, Matplotlib, Seaborn, scikit-learn (Python 3).